

Wavefunction Oneness and the EPR-Bell Nonlocality Question

Thomas Lock

thomas.lock.research@gmail.com

March 2026

Abstract

The Einstein–Podolsky–Rosen argument raised the question of whether the quantum-mechanical description of physical reality is complete [1], while Bell later showed that no local hidden-variable completion preserving a standard separability structure can reproduce the correlations predicted by quantum mechanics [2]. The claim of the present paper is narrower: the EPR–Bell nonlocality discussion may remain conceptually incomplete if it assumes separability too early.

Standard quantum mechanics already treats many multi-constituent systems as physically unified wholes described by a single composite wavefunction, including atomic bound states, multi-electron systems, molecular orbitals, and hadronic states [3, 4, 5, 6]. Interference experiments with increasingly large molecules further show that quantum oneness is not destroyed by spatial extension alone [7, 8, 9]. These accepted cases support a simple physical lesson: multiple constituents need not always be treated as independently sufficient local realities when a preserved composite state remains the relevant predictive object.

On that basis, this paper advances a conjecture of preserved composite wavefunction oneness. The conjecture is that the EPR–Bell pair may, prior to state resolution, belong to the same ontological category as other accepted composite quantum systems. If so, the pair need not be understood first as two independently complete distant entities later forced into correlation, but as one preserved composite state whose support spans separated regions and whose distinct projected outcomes become fully definite only upon resolution of the whole state. The proposal is therefore not that Bell is wrong, but that the nonlocality puzzle may be sharpened by premature separability.

Contents

1	Introduction	2
2	EPR, Bell, and the Exact Opening That Remains	3
2.1	Bell-Critical Caution	4
2.2	The Distinctive Aim of the Present Paper	5
3	Composite Oneness in Accepted Quantum Theory	5
3.1	Historical and Conceptual Precedent	5
3.2	Four Core Physical Examples	6
3.3	Interference and the Survival of Oneness Across Extension	7
3.4	Why Scale Alone Does Not Decide the Ontology	7
3.5	What These Cases Jointly Establish	8

4	Conjecture: Preserved Composite Wavefunction Oneness	9
4.1	Statement of the Conjecture	9
4.2	What “Preserved” Means	9
4.3	Constituents Without Independent Sufficiency	10
4.4	A Working Criterion for Physical Oneness	10
4.5	Why the Conjecture Matters	10
4.6	Immediate Consequence for the EPR–Bell Case	11
5	The EPR–Bell Pair as a Preserved Composite State	11
5.1	The Usual Starting Picture	11
5.2	The Composite Alternative	11
5.3	The Meaning of the Edge	12
5.4	Separated Support Does Not Yet Imply Two Physical Objects	12
5.5	Measurement from the Perspective of the Preserved Whole	13
5.6	Bell Factorization and Whole-State Description	13
5.7	The Central Reclassification	14
6	Collapse, Resolution, and the Nonlocality Question	14
6.1	Whole-State Resolution Rather Than Primitive Partial Collapse	14
6.2	Distinct Outcomes as Projections of the Resolved Whole	15
6.3	Why This Is Not Local Message Passing	15
6.4	The Nonlocality Question Reframed	15
6.5	Premature Separability	16
6.6	What This Reframing Explains	16
6.7	What Remains Open	17
7	Conclusion	17

1 Introduction

The papers of Einstein, Podolsky, and Rosen, and later Bell, remain among the most consequential logical interventions in the history of quantum theory [1, 2]. EPR did not challenge the predictive success of quantum mechanics. Their question was narrower and more foundational: whether the wavefunction can be regarded as a complete description of physical reality [1]. Bell then showed that no local hidden-variable completion preserving a standard separability structure can reproduce the correlations predicted by quantum mechanics [2]. Taken together, these two interventions did not close the foundational problem. They sharpened it.

The present paper offers a conjectured reply to that sharpened question. Its central claim is that the EPR–Bell nonlocality discussion may remain conceptually incomplete if it assumes separability too early. More specifically, this paper argues that standard quantum mechanics already licenses a physically meaningful category that has not been given its full weight in the EPR–Bell discussion: preserved composite wavefunction oneness. In many accepted cases, multiple constituents are not treated as independently sufficient physical objects, but as one unified system described by a single wavefunction. The relevant observables then belong most naturally to the whole state rather than to isolated parts considered independently.

This claim begins from ordinary quantum practice. Atomic bound states, multi-electron systems, molecular orbitals, and hadronic states are all familiar cases in which physics already treats multiple constituents as one wavefunction-defined object [3, 4, 5, 6]. Interference experiments with

increasingly large molecules further show that quantum oneness is not destroyed by spatial extension alone [7, 8, 9]. These examples do not by themselves settle the EPR–Bell question, but they establish an important physical precedent: quantum theory already recognizes real composite wholes whose predictive content belongs first to the unified state.

The conjecture advanced here is that the EPR–Bell pair may need to be reconsidered in that same light. If a shared state remains physically preserved, then the pair need not be understood first as two independently sufficient distant entities later forced into agreement by some hidden coordination. It may instead be understood as one preserved composite wavefunction whose support spans separated regions and whose distinct projected outcomes become fully definite only when the whole state is resolved. On this view, the nonlocality puzzle may be sharpened by premature separability.

The paper does not dispute Bell’s theorem or propose a hidden-variable completion. Its aim is narrower: to ask whether an already accepted category of quantum description should be brought explicitly into the EPR–Bell discussion and tested there with full seriousness.

The argument proceeds in five stages. Section 2 restates the EPR question and Bell’s constraint in the precise form needed for the present discussion. Section 3 then turns to accepted cases of composite quantum oneness, beginning with historical and conceptual precedent and continuing through four core physical examples together with interference evidence from large molecules. Section 4 formulates the conjecture of preserved composite wavefunction oneness in explicit terms. Section 5 applies that conjecture to the EPR–Bell pair itself. Finally, Section 6 examines the consequences for collapse, state resolution, and the usual reading of quantum nonlocality.

The aim throughout is not to weaken Bell, but to ask whether the EPR–Bell discussion has too often begun from the wrong ontological starting point. If standard quantum mechanics already accepts physically meaningful composite oneness elsewhere, then it is reasonable to ask whether the same category should be considered before the nonlocality problem is framed in separable terms.

2 EPR, Bell, and the Exact Opening That Remains

The starting point of the present discussion is the tension first made explicit by Einstein, Podolsky, and Rosen in 1935 [1]. Their question was not whether quantum mechanics worked empirically, but whether the wavefunction could be regarded as a complete description of physical reality. The enduring force of the EPR argument lies in the precision of its criterion:

If, without in any way disturbing a system, one can predict with certainty the value of a physical quantity, then there exists an element of physical reality corresponding to that quantity.

Under that criterion, EPR argued that if one can predict with certainty the value of a quantity for a distant system without disturbing it, then something physically real appears to correspond to that quantity whether or not the standard wavefunction assigns it a definite value prior to measurement [1]. Their challenge therefore placed three ideas into direct tension: the predictive success of quantum mechanics, the reality of definite predictable quantities, and the assumption that distant systems may be treated as separately sufficient for the purposes of physical description.

Bell later converted that conceptual tension into a mathematical constraint [2]. If one attempts to complete quantum mechanics by supplementing it with hidden variables while preserving a standard locality-style separability structure, then the resulting theory must satisfy statistical conditions that are violated by quantum mechanics and by experiment. In the usual two-party setting, let A and B denote measurement settings at two distant locations, let a and b denote the corresponding

outcomes, and let λ represent whatever additional variables are supposed to complete the description. The characteristic factorization may then be written schematically as

$$P(a, b \mid A, B, \lambda) = P(a \mid A, \lambda) P(b \mid B, \lambda). \quad (1)$$

This expression is important not merely because it is mathematically useful, but because it encodes a definite ontological picture. Once λ is specified, the pair is being treated as two outcome-bearing subsystems whose physical descriptions are locally and separately sufficient for the purposes of the correlation problem. Bell’s theorem shows that theories of this type cannot reproduce the observed quantum correlations [2].

The force of Bell’s result must be preserved exactly. The present paper does not dispute Bell’s logic, weaken his mathematical conclusion, or attempt to revive a local hidden-variable completion of this separable type. Bell remains a decisive constraint on that class of theories. What remains open, once both EPR and Bell are granted, is therefore much narrower. It is not an opening for a naive hidden-variable recovery, nor an opening to deny the observed violations of Bell inequalities. It is an opening at the level of the ontological starting point. The target of the present argument is therefore not Bell’s derivation, but the physical classification of the pair to which Bell-style separability is being applied.

That opening may be stated simply. EPR ask whether something essential is missing from the wavefunction description of physical reality [1]. Bell shows that one natural way of supplying what is missing—local hidden variables attached to separable parts—does not work [2]. The present paper asks whether another possibility remains: namely, that the measured system may have been assigned the wrong ontological category from the beginning. More specifically, the conjecture explored here is that the EPR–Bell pair, while its shared state remains preserved, may not be best described as two independently sufficient distant objects at all. It may instead belong to the same already accepted class as atoms, multi-electron states, molecular orbitals, and hadrons: composite systems whose physically complete description belongs first to one preserved wavefunction of the whole.

This is the precise point at which the present paper enters the historical conversation. EPR correctly identified a genuine incompleteness tension [1]. Bell correctly demonstrated that local separable hidden-variable completions fail [2]. The present conjecture proposes that a missing concept in that discussion may be the already accepted physical reality of preserved composite wavefunction oneness. The argument that follows therefore does not replace the original question. It attempts to sharpen its ontological starting point.

2.1 Bell-Critical Caution

At this point an important caution must be stated. Prior criticism of holist or nonseparability-based responses to Bell has emphasized that one cannot dissolve Bell’s theorem merely by replacing the language of separate parts with the language of the whole. [10, 11, 12] Bell’s result is a mathematical constraint on theories satisfying a precise locality-style separability structure. A merely verbal appeal to wholeness does not by itself remove that constraint.

The present paper accepts this caution fully. [10, 12] It does not claim that Bell’s theorem disappears once one invokes oneness. It does not claim that generic holism, by itself, answers Bell. Bell’s theorem remains valid as a theorem about local hidden-variable completions of the separable kind.

The present paper’s claim is narrower. It asks whether the EPR–Bell pair should be classified, prior to resolution, as the kind of system to which Bell-style separable factorization should be applied in the first place. That is the exact pressure point. The argument does not challenge Bell’s

mathematics; it challenges the ontological starting point from which Bell’s assumptions are taken to describe the system.

2.2 The Distinctive Aim of the Present Paper

The present paper therefore differs from generic holism or nonseparability discussions in three specific ways.

First, it does not begin from abstract ontology alone. It begins from accepted physical examples already standard in atomic, molecular, and hadronic quantum mechanics.

Second, it employs a stricter notion than ordinary structural connectedness. The term *oneness* in this paper does not mean mere cohesion, aggregate structure, or mechanical connectedness. A rock, crystal, or rigid body may possess strong internal structure while still permitting largely independent examination of its parts. By contrast, *wavefunction oneness* refers to a preserved composite state whose physically complete description belongs to the whole rather than to independently sufficient local components.

Third, the present argument is not offered as a broad reinterpretation of all of quantum mechanics at once. It is offered as a direct logical reply to the EPR–Bell nonlocality question. Its proposal is that an already accepted category of physical description — preserved composite wavefunction oneness — may be the missing conceptual premise in that discussion.

3 Composite Oneness in Accepted Quantum Theory

Before applying the present conjecture to the EPR–Bell case, it is necessary to identify a point that is already embedded in ordinary quantum practice. Standard quantum mechanics does not treat every multi-constituent system as a mere aggregate of independently sufficient local objects. In many accepted cases, the predictive object is a composite wavefunction of the whole, and the relevant observables belong first to that unified state. The present section gathers the historical, conceptual, and physical precedent for that claim.

3.1 Historical and Conceptual Precedent

The idea that the whole may enjoy a form of priority over the parts is not foreign to the history of quantum theory. Schrödinger himself, in the wake of the EPR discussion, emphasized that entangled or correlated systems cannot always be understood by assigning complete independent descriptions to their components taken in isolation [13, 14]. Later philosophical and foundational discussions of holism and nonseparability developed related themes in different language, asking whether quantum theory permits physically meaningful wholes whose properties are not reducible to independently complete local parts [15, 16, 11, 17, 10, 12].

The present paper is continuous with that broad line of thought, but its aim is narrower and more concrete. It does not attempt to settle the full philosophical literature on holism. Instead, it asks a more direct physical question: where does standard quantum mechanics already treat multiple constituents as one wavefunction-defined object, and what lesson should be drawn from those accepted cases when approaching the EPR–Bell pair? The argument therefore begins not with abstract metaphysics, but with familiar quantum systems.

It is also important to distinguish wavefunction oneness from more ordinary forms of wholeness. A rock, crystal, or rigid body may possess strong internal structure while still permitting largely independent examination of its parts. Such systems may be unified in a structural or macroscopic

sense without requiring that the physically complete predictive description belong first to one irreducible state of the whole. The kind of oneness relevant here is different. Preserved composite wavefunction oneness refers to cases in which the physically meaningful description belongs primarily to the composite quantum state itself, and in which the parts are not independently sufficient at that descriptive level.

3.2 Four Core Physical Examples

One of the earliest decisive successes of wave mechanics was the hydrogen atom. In the standard treatment, hydrogen is not handled as a proton and electron each carrying independently sufficient physical descriptions for the purposes of spectroscopy. It is treated as one bound quantum system with one Hamiltonian, one wavefunction, and one spectrum. In the usual separation of variables, the stationary states are written as

$$\Psi_{n\ell m}(r, \theta, \phi) = R_{n\ell}(r)Y_{\ell m}(\theta, \phi), \quad (2)$$

with the ground-state orbital given by

$$\Psi_{100}(r) = \frac{1}{\sqrt{\pi a_0^3}} e^{-r/a_0}. \quad (3)$$

The important point is not merely formal. The relevant observables—energy levels, transition frequencies, and orbital structure—belong to the atom as a whole. Hydrogen already shows that a spatially extended bound system may be treated as one physical object described by one wavefunction.

Helium strengthens the point. Here the accepted description is explicitly multi-electron and composite. In the simplest orbital picture, the spatial part of the ground state may be written as

$$\Psi_{\text{spatial}}(r_1, r_2) = \phi_{1s}(r_1)\phi_{1s}(r_2), \quad (4)$$

while the spin part is the singlet

$$\chi_{\text{singlet}} = \frac{1}{\sqrt{2}} (|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle). \quad (5)$$

Thus the full state takes the form

$$\Psi_{\text{He}}(r_1, r_2) = \phi_{1s}(r_1)\phi_{1s}(r_2) \frac{1}{\sqrt{2}} (|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle). \quad (6)$$

The physically meaningful description belongs to the two-electron state as a whole, not to two independently sufficient electronic descriptions considered first and only later combined. Helium therefore provides an accepted example in which multiplicity of constituents does not eliminate physical oneness [3].

The same pattern appears in molecular bonding. In the Heitler–London treatment, and later in molecular-orbital language, the state of a bonded system is described by a wavefunction extending across more than one nucleus [4]. For the hydrogen molecule, a symmetric bonding form may be written schematically as

$$\Psi_{\text{bond}} \propto \phi_A(1)\phi_B(2) + \phi_B(1)\phi_A(2). \quad (7)$$

Here again the important point is conceptual rather than merely algebraic. The bonded molecule is not treated as two already complete atomic realities with an external bookkeeping relation added

afterward. Its physically relevant identity is carried by the delocalized composite state itself. Bonding is one of the earliest and clearest demonstrations that a quantum state can define a unified object across more than one primitive center.

Hadrons provide an even sharper example, and in the present context they may be the strongest precedent of all. In the quark model, baryons and mesons are not treated as independently sufficient free quarks merely traveling together. They are treated as confined composite quantum objects whose state is defined only at the level of the hadron as a whole [5, 6]. Confinement is crucial here because it blocks any straightforward move from constituent structure to independently sufficient observable parts. Whatever the underlying constituent content may be, the physically realized observables—mass, spin, charge, decay channels, and scattering signatures—belong to the composite hadronic state. In that sense, hadrons show in especially strong form that physical description may belong first to the unified state rather than to isolated constituents. For the purposes of the present paper, they therefore provide the strongest accepted precedent for denying primitive independent sufficiency to the parts while preserving the physical reality of the whole.

These examples are not peripheral curiosities. They belong to the historical core of quantum theory itself. At the levels of atomic physics, many-electron structure, molecular chemistry, and strong-interaction phenomenology, standard practice already recognizes composite wavefunction-defined wholes.

3.3 Interference and the Survival of Oneness Across Extension

Even the familiar double-slit experiment already illustrates the essential point in miniature. A quantum particle passing through a two-slit arrangement is not ordinarily treated as two independently sufficient localized objects, one at each slit, later forced into agreement. It is treated as one wavefunction with distributed support whose interference pattern reveals that the physically relevant predictive object remained the unified state of the whole experimental configuration. In that sense, double-slit interference provides the simplest intuitive example of quantum oneness across separated support. The large-molecule interference results discussed below matter because they show that this principle is not confined to the most elementary cases, but survives for composite systems of increasing size and complexity.

The preceding examples are all bound or internally organized systems, but one might still wonder whether their unity depends on small scale alone. Interference experiments with large molecules show that it does not. Matter-wave interference has been observed for fullerene molecules, for increasingly large organic compounds, and for molecular systems extending far beyond the scale at which one might once have expected quantum behavior to reduce trivially to localized classical parts [7, 8, 9].

The force of these results for the present paper is straightforward. They do not by themselves prove the present conjecture about the EPR–Bell pair. But they do show that quantum oneness is not destroyed merely because a system is extended, composite, or large by familiar microscopic standards. A composite wavefunction may remain the physically relevant predictive object even when its support spans a substantial region. Spatial extension therefore cannot serve as a primitive criterion for denying oneness.

3.4 Why Scale Alone Does Not Decide the Ontology

This point may be stated more sharply. A wavefunction is not, in the first instance, tagged with a primitive geometric rule saying that below some length scale it counts as one object and above that scale it must count as many. Quantum theory instead asks what the preserved predictive state is,

what observables belong to it, and whether the physical description remains attached to the whole configuration. Scale may matter dynamically, but scale alone does not settle ontology.

That is why the familiar examples matter so much. Hydrogen occupies a nontrivial region yet is one atom. Helium is spatially extended and internally structured yet is one atomic system. Molecules extend across more than one center yet are treated as one bonded object. Large molecules can still interfere as one quantum system. Hadrons may involve multiple constituents yet are not treated as freely separable physical objects within the relevant predictive description. In every case, multiplicity and extension fail to force primitive separability.

This also clarifies an important distinction for what follows. Support across a region is not the same thing as influence across a region. A wavefunction may extend over separated or distributed support without this meaning that one already has many independently sufficient physical objects exchanging later signals. The existence of extended support does not by itself decide between one preserved whole and several separate realities.

The resulting working principle is scale-independent: no fixed geometric scale alone determines when preserved composite wavefunction oneness must fail.

3.5 What These Cases Jointly Establish

The lesson of the accepted cases can now be stated in a form directly relevant to the present paper. Standard quantum mechanics already licenses systems in which

1. multiple constituents are present,
2. the state may extend across a nontrivial region or across more than one center,
3. the predictive description belongs first to a single composite wavefunction, and
4. the relevant observables belong more naturally to the whole than to isolated parts treated as independently sufficient.

This is the key precedent. The point is not merely that composite description is mathematically convenient. The point is that composite description is often taken to represent the physically real predictive object. In that sense, the whole is not a decorative summary of already complete parts.

In these accepted cases, the whole is greater than the isolated parts in a specifically quantum sense. This does not mean that the constituents vanish or become fictitious. It means that, at the relevant descriptive level, the physically primary object is the composite state itself, while the parts are no longer independently sufficient to exhaust the predictive content of the system. The whole is what makes the parts physically intelligible in the first place within the accepted quantum description.

The present argument therefore does not introduce composite oneness as a speculative novelty. It identifies it as an already functioning category of standard quantum theory and asks whether the EPR–Bell pair may belong, prior to state resolution, to that same ontological class. If that possibility is to be assessed fairly, it must be assessed against the background of these accepted cases rather than against an unexamined presumption that spatially distributed systems are automatically many from the start.

The guiding question is therefore not simply how far apart the parts may be before they stop being one thing. The deeper question is under what conditions a preserved composite wavefunction continues to define one physically meaningful object. That is the question to which the conjecture of the next section is addressed.

4 Conjecture: Preserved Composite Wavefunction Oneness

The preceding section established a limited but important point: standard quantum mechanics already recognizes physically meaningful composite systems whose predictive description belongs first to a unified wavefunction of the whole. The present section now states the conjecture that guides the remainder of the paper.

4.1 Statement of the Conjecture

The central conjecture of this paper may be stated as follows:

Conjecture. When a quantum system remains physically described by a preserved composite wavefunction, the relevant physical object is the whole state rather than a collection of independently sufficient parts. In such cases, multiplicity of constituents and spatially distributed support do not by themselves require primitive separability.

Applied in the narrower context of the present paper, the conjecture becomes:

EPR–Bell Form of the Conjecture. The EPR–Bell pair may, prior to state resolution, belong to the same ontological category as other accepted composite quantum systems: one preserved composite wavefunction whose support spans separated regions, but whose physically relevant description still belongs first to the whole.

The claim is intentionally narrow. It does not deny the existence of constituents, erase spatial separation, or challenge Bell’s theorem. Its point is simply that quantum theory already contains cases in which the physically primary description belongs to a preserved whole, and that the EPR–Bell pair may need to be reconsidered under that same category.

4.2 What “Preserved” Means

The term *preserved* is essential. The present conjecture does not apply to any arbitrary mathematical tensor product, nor to every formally composite system. It applies only when the physically relevant predictive description still belongs to a unified state of the whole. In that sense, preservation means that the composite wavefunction remains the correct state-bearing object for the purposes of the observables under discussion.

Formally, if a system of two components is described by a state Ψ_{AB} , then the question is not merely whether A and B can be named as distinguishable loci or constituents. The question is whether, for the physical situation at hand, the state of the system is still properly carried by

$$\Psi_{AB}, \tag{8}$$

rather than by two independently sufficient realized descriptions

$$\psi_A \quad \text{and} \quad \psi_B \tag{9}$$

taken as primitive prior to resolution of the shared state.

In that sense, preservation means that the whole remains the predictive object. The wavefunction of the whole has not yet been replaced, for the relevant physical purpose, by separately sufficient local state assignments that exhaust the physical content. One may speak of distinguishable regions, measurement settings, or projected outcomes, but those do not automatically imply that the underlying predictive object has already fragmented into ontologically independent realized pieces. The conjecture therefore applies only so long as the composite wavefunction remains physically preserved as the relevant predictive object, rather than having become broken, environmentally disrupted, or otherwise replaced by independently sufficient realized descriptions.

4.3 Constituents Without Independent Sufficiency

This distinction between *constituency* and *independent sufficiency* is the heart of the conjecture. A composite system may certainly contain distinguishable constituents, centers, or regions of support. Hydrogen contains quarks, a proton and an electron. Helium contains two electrons and a nucleus. A molecule contains multiple nuclei and electrons. A hadron contains multiple quark degrees of freedom. Yet in the accepted quantum description, these constituents are not thereby treated as independently sufficient physical objects for the observables that define the state.

The same logical distinction is being proposed here for the EPR–Bell pair. Two regions of support, or two measurement wings, do not automatically imply two already complete physical objects in the ontological sense relevant to the correlation problem. What matters is whether the physical description is still being carried by the preserved composite state. If it is, then the parts are present as constituents of the whole, but not yet as independently sufficient realized objects whose full physical identity has been fixed apart from the state of the whole.

This point may be expressed schematically as follows. Let the composite state be written

$$\Psi_{AB} \neq \psi_A \otimes \psi_B \tag{10}$$

in the sense relevant to physical realization prior to state resolution. Equation (10) does not deny that product decompositions may be used in formal constructions or post-measurement descriptions. Its point is ontological rather than merely algebraic: the physically relevant object need not be two already complete local realizations simply because two loci are available for later projection.

4.4 A Working Criterion for Physical Oneness

The conjecture may now be sharpened into a working criterion. A quantum system should be treated as exhibiting preserved composite wavefunction oneness when the following conditions hold:

1. the physically relevant predictive description is carried by a composite state of the whole;
2. the observables of interest belong first to that whole-state description rather than to independently sufficient local realizations taken as primitive;
3. spatial distribution or multiplicity of constituents does not, by itself, destroy the physical unity of the state; and
4. distinct definite outcomes arise only through resolution, projection, or decomposition of the whole state rather than being primitively complete beforehand.

This criterion is intentionally modest. It does not claim to solve every interpretive question about quantum ontology. It functions only as a disciplined way to identify when standard quantum practice already treats a composite wavefunction as physically primary. The conjecture of the present paper is that the EPR–Bell case may satisfy that criterion more naturally than is usually acknowledged.

4.5 Why the Conjecture Matters

The importance of the conjecture is not that it introduces a new exotic structure into quantum theory. Its importance is that it shifts the order of description. In the usual separability-first picture, one begins with two distant systems and then asks how their correlated outcomes can be

reconciled with locality. In the present conjectural picture, one begins instead with one preserved composite state and asks how distinct local outcomes emerge upon resolution of that whole.

That is a different ontological starting point, and the difference is not trivial. If the relevant physical object is one preserved whole, then the correlation problem is not initially the problem of coordinating two already complete distant realities. It is the problem of how one shared state yields distinct projected outcomes when resolved under separated measurement conditions. The emphasis therefore shifts from post hoc coordination between parts to pre-resolution unity of the predictive object.

4.6 Immediate Consequence for the EPR–Bell Case

The immediate consequence is straightforward. If the conjecture is granted even as a serious possibility, then the EPR–Bell pair cannot simply be assumed from the outset to be two independently sufficient distant objects whose relation is secondary. The ontological burden of proof changes direction. One must first ask whether the shared state remains physically preserved as a whole.

Only after that question has been answered is it legitimate to decide whether Bell-style separability assumptions are being applied at the right descriptive level. The next section therefore turns directly to the EPR–Bell pair itself and asks what changes when it is approached, not first as two completed distant objects, but as one preserved composite wavefunction whose distinct local outcomes emerge only upon resolution.

5 The EPR–Bell Pair as a Preserved Composite State

The preceding sections have prepared the central question of this paper. If standard quantum mechanics already recognizes physically meaningful composite wholes in other accepted cases, then the EPR–Bell pair must be examined to determine whether it may belong, prior to state resolution, to that same ontological category. The present section develops that application directly.

5.1 The Usual Starting Picture

In the standard presentation of the EPR–Bell problem, one begins with two spatially separated measurement wings, typically associated with two observers, two devices, or two distant regions of support. The pair is then treated, at least for purposes of the locality problem, as though the physically relevant question concerns two distinct systems whose outcomes must be coordinated across separation. From that starting point, Bell’s theorem becomes a constraint on whether any local separable completion can reproduce the observed correlations [2].

That framing is powerful, but it may also import an ontological assumption too quickly. It encourages one to think of the pair first as two independently sufficient distant entities and only afterward as correlated. The present paper asks whether that order should be reversed. The issue is not whether two measurement wings exist, nor whether distinct outcomes are eventually recorded at separated sites. The issue is whether those later facts justify treating the pre-resolution physical object as already ontologically divided into two complete local realities.

5.2 The Composite Alternative

The alternative proposed here is that the EPR–Bell pair may, prior to state resolution, be more accurately regarded as one preserved composite state. In that description, the pair is not first understood as two complete physical objects later coordinated by an additional relation. It is

understood instead as one wavefunction-defined whole whose support spans separated regions and whose later projected outcomes are extracted from that whole-state description.

To make the point schematic, let the shared state be written as

$$\Psi_{AB}. \quad (11)$$

The present conjecture is that, so long as this shared state remains physically preserved, the relevant predictive object is

$$\Psi_{AB}, \quad (12)$$

not two already sufficient realized states ψ_A and ψ_B taken as ontologically primary. In that sense, the pair is not being denied multiplicity of support. It is being denied primitive separability as the automatic starting point.

This may be expressed again in the form

$$\Psi_{AB} \neq \psi_A \otimes \psi_B \quad (13)$$

where the intended force of the expression is ontological rather than merely formal. The point is not that tensor-product language is never mathematically available, but that the physically relevant pre-resolution object need not be interpreted as two independently complete local realizations simply because later measurements occur in distinct places.

5.3 The Meaning of the Edge

To describe this more intuitively, it is useful to speak of the pair not merely as A and B , but as

$$A + \text{edge} + B. \quad (14)$$

The language of the *edge* is not meant to introduce a new force, signal, or medium. It is purely schematic terminology for treating the relation between the two wings as constitutive of the whole-state description rather than as an externally added link between already complete local objects. It is meant to prevent the relation between the two wings from being treated as an afterthought.

In that sense, the edge is not logically disposable. If the EPR–Bell pair is physically complete only as a preserved whole, then the shared relation is part of what the object *is*. The edge is therefore not a mere mathematical convenience joining two otherwise complete local entities. It belongs to the physical completeness of the composite description itself.

The point of this language is conceptual economy. A bonded molecule is not well described as atom A , plus atom B , plus an externally attached bookkeeping relation. The bonded state is the object. Likewise, if the EPR–Bell state remains physically preserved, then the connection between the two wings is not something added after the fact. It belongs to the identity of the composite state itself.

5.4 Separated Support Does Not Yet Imply Two Physical Objects

At this point the earlier distinction between support and influence becomes decisive. The fact that the state has support in separated regions does not by itself show that one already has two independently sufficient physical objects. Standard quantum theory already contains accepted cases in which a wavefunction extends across multiple centers or across a substantial region while still counting as one physically relevant state. Section 3 established that point from familiar examples.

The same caution should therefore be applied here. The existence of two measurement wings, or of two possible local projections, does not settle the ontological question in advance. It tells us

where state resolution may later become manifest. It does not yet tell us whether the pre-resolution object was one preserved whole or many independently complete realities.

This is precisely where the present paper departs from the usual reading. The usual reading begins with many and asks how correlation is possible. The present conjecture begins with one preserved whole and asks how distinct local outcomes emerge upon resolution. That is a genuine shift in explanatory order.

5.5 Measurement from the Perspective of the Preserved Whole

If the EPR–Bell pair is treated as one preserved composite state, then measurement must be described accordingly. A measurement on one wing is not, in the first instance, the revelation of a property belonging to an already independently complete local object. It is an interaction through which the preserved whole is forced into definite projected structure under a specific experimental arrangement.

That does not mean the distant wing is physically unreal, nor that no distinction exists between local settings. It means that the full physical description is not exhausted by assigning primitive local completeness before the shared state has been resolved. The measurement acts on the composite state as encountered from one wing under one setting, and the eventual definiteness of outcomes belongs to the resolved structure of that whole.

In this sense, one may say that the local apparatus addresses the whole through a local interaction. This is already familiar in less controversial composite systems. One probes a molecule, an atom, or a many-body state through localized interactions, yet what is being probed is not thereby reduced to a primitive collection of independently sufficient local realities. The present conjecture simply asks whether the EPR–Bell state should be treated the same way prior to resolution.

5.6 Bell Factorization and Whole-State Description

The importance of this shift becomes clearest when set beside Bell’s factorization condition. In the separability-first picture, one asks whether the joint probabilities can be written as

$$P(a, b \mid A, B, \lambda) = P(a \mid A, \lambda) P(b \mid B, \lambda), \quad (15)$$

with λ specifying the relevant completing variables [2]. This form presupposes that, once the underlying state of affairs is fixed by λ , the two wings admit separately sufficient outcome assignments conditioned only on local settings.

But if the relevant pre-resolution object is instead one preserved composite state, then the natural conditional structure is not the same. The more basic description is

$$P(a, b \mid A, B, \Psi_{AB}), \quad (16)$$

where the joint outcome structure is conditioned on the shared state of the whole. On this view, the correlation is not something produced by stitching together two already complete local descriptions. It belongs to the predictive structure of the preserved composite state itself.

This does *not* refute Bell. Bell’s theorem remains fully valid as a theorem about local separable completions of the relevant type. The present point is narrower. It is that Bell’s factorization condition may be applied only after one has settled the ontological status of the physical object under discussion. If the EPR–Bell pair has been assumed to be primitively separable at the wrong stage of description, then the nonlocality puzzle may have been posed from an ontologically premature starting point.

5.7 The Central Reclassification

The proposal of this paper may now be stated in its final applied form. Prior to state resolution, the EPR–Bell pair need not be classified first as two independently sufficient distant physical objects. It may instead be more faithfully classified as one preserved composite wavefunction whose support spans separated regions and whose distinct local outcomes become definite only through resolution of the whole state.

This is the central reclassification on which the argument depends. It does not deny that there are two wings, two settings, or two eventual local records. It says only that these features do not by themselves settle the ontology of the pre-resolution state. The mere existence of later local definiteness is not yet a warrant for primitive separability at the earlier descriptive stage.

The importance of this reclassification is methodological as much as ontological. If it is mistaken, then the ordinary separability-first reading remains intact. But if it is correct, even in part, then the nonlocality problem has been posed too late in the conceptual sequence. One must first decide what kind of physical object the shared state is before asking how its later local outcomes are to be understood across separation.

The next section therefore does not reopen the examples or restate the conjecture. It asks what follows once the pair is treated, prior to resolution, as one preserved composite whole. In that setting, collapse is no longer described first as coordination between two already complete distant realities, but as the resolution of a single shared predictive state into definite local projections.

6 Collapse, Resolution, and the Nonlocality Question

If the EPR–Bell pair is treated, prior to measurement, as a preserved composite wavefunction, then the meaning of collapse must be reconsidered accordingly. The present section develops that consequence and then turns to the usual nonlocality question. The guiding idea is simple: if the predictive object is one preserved whole, then the emergence of distinct outcomes should be described first as the resolution of that whole rather than as the coordination of two already complete distant realities.

6.1 Whole-State Resolution Rather Than Primitive Partial Collapse

In the usual picture, measurement on one wing is often imagined as though it acts upon one member of an already physically complete pair and then somehow constrains what may be found on the other wing. That picture is natural if one begins from primitive separability. But it is no longer the most natural picture if the pre-resolution object is one preserved composite state.

On the conjecture advanced here, collapse is better described as *whole-state resolution*. The shared state

$$\Psi_{AB} \tag{17}$$

is the physically relevant predictive object prior to outcome definiteness. When one measurement interaction occurs under a chosen setting, it does not merely reveal a property of an already independently sufficient local object. It contributes to the forced resolution of the preserved whole into definite projected structure. In that sense, the local interaction is real and localized, but the state whose definiteness is being resolved is not merely local.

This is why the distinction developed earlier is so important. The existence of local apparatus, local settings, and local records does not by itself imply that the pre-measurement ontology was already composed of independently complete local objects. The present conjecture instead proposes

the sequence

$$\text{preserved composite state} \longrightarrow \text{whole-state resolution} \longrightarrow \text{definite local projections.} \quad (18)$$

The order matters. The distinct local outcomes are not primitively complete and then correlated afterward. They arise as projections of the resolved whole.

6.2 Distinct Outcomes as Projections of the Resolved Whole

Once the shared state is resolved, one indeed obtains distinct outcomes at distinct locations. Nothing in the present argument denies that experimental facts are recorded in separated wings, or that those records are definite in the ordinary operational sense. What changes is the ontological order in which those facts are understood.

Let the possible outcome channels be represented schematically by basis states $\{|a_i\rangle_A\}$ and $\{|b_j\rangle_B\}$. Then the shared state may be written in the familiar form

$$\Psi_{AB} = \sum_{i,j} c_{ij} |a_i\rangle_A |b_j\rangle_B. \quad (19)$$

The present point is not to alter this mathematics, but to interpret its physical role differently. Prior to resolution, Equation (19) represents one composite predictive state. After interaction with the measurement arrangement, definite outcomes correspond to the realized projection structure of that whole. The local results are therefore not treated as primitive independent realities later tied together by an additional explanatory bridge. They are treated as coordinated projections emerging from one resolved state.

This is the sense in which the whole remains explanatorily primary even though the final records are local. The locality of the records does not force primitively local completeness at the pre-resolution stage. It shows only where and how the whole state becomes experimentally articulated.

6.3 Why This Is Not Local Message Passing

This shift in perspective also clarifies what the present paper is *not* proposing. If the EPR–Bell pair is one preserved composite state, then the correlation between outcomes need not be explained by any signal sent from one completed local object to another at the moment of measurement. No such message-passing picture is required by the present conjecture.

The reason is straightforward. If the two outcomes are projections of one resolved whole, then their correlation belongs to the structure of that whole-state resolution rather than to an exchange between two already independently sufficient distant realities. The issue is therefore not how one wing informs the other after the fact. The issue is how one shared state yields distinct but correlated local projections when resolved under separated conditions.

This point parallels accepted composite systems elsewhere in quantum theory. When one probes a molecule or many-body state through a localized interaction, one does not usually imagine that the measured region must send a classical-style message to the rest of the system in order for the state description to remain unified. Rather, one accepts that the localized interaction addresses a state whose physically relevant description belongs to the whole. The present conjecture asks whether the EPR–Bell case should be treated in the same general way prior to state resolution.

6.4 The Nonlocality Question Reframed

We may now return to the standard nonlocality question. In the usual reading, the experimental violation of Bell inequalities is taken to show that no local separable completion of the relevant kind

can account for the correlations [2]. That conclusion remains intact. But the present conjecture suggests that one further question should be asked before the full conceptual weight of “nonlocality” is assigned.

That further question is whether the system under discussion was ontologically separable in the first place at the stage where the puzzle is being posed. If the EPR–Bell pair has already been assumed to consist of two independently sufficient distant objects prior to resolution, then the observed correlations naturally appear as a problem of cross-distance coordination. But if the pair is instead one preserved composite state, then the primary conceptual task changes. One no longer asks first how two complete distant realities remain correlated. One asks how one shared state yields two definite local projections under separated measurement conditions.

This is not a semantic adjustment. It changes the explanatory burden. In the separability-first reading, nonlocality appears as the mysterious persistence of relation across ontological division. In the preserved-whole reading, the mystery is relocated. The puzzle is no longer how two primitively separate objects communicate, but how one wavefunction-defined whole resolves into distinct local definiteness without being misdescribed in advance as many.

6.5 Premature Separability

The central claim of the present paper may therefore be restated at its sharpest point: the usual nonlocality problem may be *sharpened by premature separability*. By this is meant the assumption that spatial separation of support, settings, and later records is already sufficient to force ontological separability prior to the resolution of the shared state.

The accepted composite examples reviewed earlier show that this inference is not generally licensed by quantum practice. Composite wavefunctions can remain physically primary across multiple constituents, across multiple centers, and across nontrivial spatial extension. If so, then the mere existence of two wings in the EPR–Bell arrangement does not yet decide the ontology. To infer primitive many-ness too early may be to build the nonlocality puzzle on a starting point that standard quantum theory elsewhere does not require us to accept.

This is the precise sense in which the present paper reopens the EPR–Bell question from within quantum theory rather than against it. The proposal is not that Bell is wrong. It is that Bell’s theorem may be describing the failure of local separable completions for a system that should first have been examined as a preserved composite whole.

6.6 What This Reframing Explains

If the conjecture is granted, even provisionally, then several conceptual gains follow. First, the correlation between outcomes is no longer treated as something that must be added to two already complete local realities. It belongs instead to the shared predictive structure of the preserved whole. Second, collapse is no longer pictured primarily as an action on one part with mysterious implications for another distant part. It is described as the resolution of one composite state into definite local projections. Third, the term “nonlocality” no longer has to bear the full explanatory burden by default, because the underlying ontological classification of the system has been reopened.

The point of this reframing is therefore not to replace Bell’s theorem or to offer a finished theory of measurement. Its value is narrower and more disciplined. Before the EPR–Bell case is described as a problem of relation between two independently complete distant objects, one should first ask whether the pair is better understood as one preserved composite wavefunction whose later definiteness appears through whole-state resolution.

6.7 What Remains Open

These conceptual gains do not eliminate every open question. They instead clarify what still needs to be sharpened if the present conjecture is to be developed further.

The preservation criterion remains the most immediate task. The argument of this paper depends on the distinction between a genuinely preserved composite state and a merely formal composite description, but that boundary still requires more rigorous statement. The relation between the present conjecture and existing interpretive accounts of collapse also deserves fuller development, especially if the proposal is to be situated more precisely within the broader foundations literature.

A second open task concerns the probabilistic bridge to Bell-style analysis. The present paper has argued that Bell's factorization may not be physically appropriate unless the ontological status of the EPR–Bell pair has first been settled. But the exact mathematical relation between Bell-style probability factorizations and whole-state descriptions still requires more systematic treatment than has been attempted here.

These open problems do not weaken the central point of the argument. They define its present scholarly status. The claim of this paper remains limited but exact: the EPR–Bell pair may be misclassified if it is treated as primitively separable too early. If that possibility is genuine, then the nonlocality question remains important, but it must be asked from a more careful starting point.

7 Conclusion

The argument of this paper has been narrow, but its implications are substantial. Einstein, Podolsky, and Rosen asked whether the quantum-mechanical description of physical reality is complete [1]. Bell later showed that no local hidden-variable completion preserving a standard separability structure can reproduce the correlations predicted by quantum mechanics [2]. Those results remain intact. The present paper has instead asked whether the EPR–Bell discussion may still be conceptually incomplete if separability is assumed too early.

The case developed here has proceeded from within standard quantum practice. Quantum theory already recognizes many accepted systems in which multiple constituents are treated, not as independently sufficient physical objects, but as one wavefunction-defined whole. Atomic bound states, multi-electron systems, molecular orbitals, hadronic states, and interference experiments with increasingly large molecules all show that multiplicity of constituents and spatial extension do not by themselves destroy physical oneness [3, 4, 5, 6, 7, 8, 9]. These examples do not settle the EPR–Bell question by analogy alone, but they do establish an already accepted ontological category: preserved composite wavefunction oneness.

On that basis, this paper advanced a conjecture. The conjecture is that the EPR–Bell pair may, prior to state resolution, belong to the same ontological category as other accepted composite quantum systems. If so, then the pair need not be understood first as two independently sufficient distant entities later forced into correlation. It may instead be understood as one preserved composite wavefunction whose support spans separated regions and whose distinct local outcomes become fully definite only through resolution of the whole state.

From that starting point, the usual reading of nonlocality is altered. Bell's theorem still rules out local separable hidden-variable completions of the relevant kind [2]. But the present argument suggests that the conceptual burden of nonlocality may be sharpened by premature separability: the assumption that separated support, settings, and later records are already sufficient to force ontological division prior to resolution of the shared state. If that assumption is mistaken, then the primary question is not how two already complete distant realities remain coordinated, but how one

preserved composite state yields distinct correlated local projections when resolved under separated conditions.

The contribution of the paper is therefore ontological rather than dynamical. It is logical rather than programmatic: it advances a conjectural reclassification of the EPR–Bell pair within existing quantum practice rather than a new hidden-variable mechanism or predictive departure from standard quantum mechanics. The preservation criterion still requires further sharpening, and the probabilistic bridge between Bell-style factorizations and whole-state descriptions deserves fuller mathematical development. But those open tasks do not alter the central point: before the EPR–Bell case is framed as a relation between two independently complete distant objects, one should first ask whether the shared state is more faithfully understood as a preserved composite whole.

The EPR–Bell discussion has often been framed as though its decisive ontological step has already been taken before the real argument begins. The present paper suggests otherwise. If standard quantum mechanics already accepts physically meaningful composite oneness elsewhere, then it is reasonable to ask whether the same category should be taken seriously here as well. In that sense, the paper does not close the EPR–Bell question. It reopens it at what may be the more faithful starting point.

References

- [1] Albert Einstein, Boris Podolsky, and Nathan Rosen. Can quantum-mechanical description of physical reality be considered complete? *Physical Review*, 47(10):777–780, 1935.
- [2] J. S. Bell. On the einstein podolsky rosen paradox. *Physica Physique Fizika*, 1(3):195–200, 1964.
- [3] Egil A. Hylleraas. Neue berechnung der energie des heliums im grundzustande, sowie des tiefsten terms von ortho-helium. *Zeitschrift für Physik*, 48(7–8):469–494, 1928.
- [4] W. Heitler and F. London. Wechselwirkung neutraler atome und homöopolare bindung nach der quantenmechanik. *Zeitschrift für Physik*, 44(6–7):455–472, 1927.
- [5] Murray Gell-Mann. A schematic model of baryons and mesons. *Physics Letters*, 8(3):214–215, 1964.
- [6] George Zweig. An $su(3)$ model for strong interaction symmetry and its breaking. *CERN Report*, 8182/TH.401 and 8419/TH.412, 1964. Originally circulated as CERN preprints.
- [7] Markus Arndt, Olaf Nairz, Julian Vos-Andreae, Claudia Keller, Gerbrand van der Zouw, and Anton Zeilinger. Wave–particle duality of c_{60} molecules. *Nature*, 401:680–682, 1999.
- [8] Stefan Gerlich, Sandra Eibenberger, Mathias Tomandl, Stefan Nimmrichter, Klaus Hornberger, Paul J. Fagan, Jens Tüxen, Marcel Mayor, and Markus Arndt. Quantum interference of large organic molecules. *Nature Communications*, 2:263, 2011.
- [9] Yaakov Y. Fein, Philipp Geyer, Patrick Zwick, Filip Kiałka, Sebastian Pedalino, Marcel Mayor, Stefan Gerlich, and Markus Arndt. Quantum superposition of molecules beyond 25 kda. *Nature Physics*, 15:1242–1245, 2019.
- [10] Joe Henson. What bell did. *Foundations of Physics*, 45(5):503–526, 2015.

- [11] Federico Laudisa. The epr argument in a relational interpretation of quantum mechanics. *Foundations of Physics Letters*, 8(4):371–387, 1995.
- [12] Darrin W. Belousek. Does quantum nonseparability answer bell’s theorem? *Foundations of Physics*, 52(6):1–30, 2022.
- [13] Erwin Schrödinger. Discussion of probability relations between separated systems. *Proceedings of the Cambridge Philosophical Society*, 31:555–563, 1935.
- [14] Erwin Schrödinger. Probability relations between separated systems. *Proceedings of the Cambridge Philosophical Society*, 32:446–452, 1936.
- [15] Paul Teller. Relational holism and quantum mechanics. *British Journal for the Philosophy of Science*, 37(1):71–81, 1986.
- [16] Don Howard. Holism, separability, and the metaphysical implications of the bell experiments. *Noûs*, 23(5):653–674, 1989.
- [17] Richard Healey. Holism and nonseparability in physics. In Edward N. Zalta and Uri Nodelman, editors, *The Stanford Encyclopedia of Philosophy*. Metaphysics Research Lab, Stanford University, 2020. Fall 2020 Edition.